

Yixiang Luo

Computational Systems Biology · AI Virtual Cells · Genome-Scale Models

vivarock233@gmail.com · vivarock233-byte.github.io
Dalian, China · 15581332700



Computational systems biology researcher experienced in strain-specific GEM reconstruction, reference-guided gap filling, FBA/pFBA, comparative genomics, and phenotype analysis. Interested in hybrid mechanistic-AI virtual cells and evidence-aware agents for metabolic-model curation.

EDUCATION

Master's Student in Biology and Medicine · Dalian Institute of Chemical Physics, Chinese Academy of Sciences 2026–Present

Primary supervisor: Prof. Yongjin Zhou · Joint-training supervisor: Asst. Prof. Feiran Li, Tsinghua University

B.Eng. · Dalian University of Technology 2022–2026
Academic rank: 2nd

Exchange Student · National Yang Ming Chiao Tung University 2023
Faculty mentor: Assoc. Prof. Yi-Hsueh Chuang

HONORS & AWARDS

- **Outstanding Graduate**, Dalian University of Technology, 2026
- **University Outstanding Student Honor**, Dalian University of Technology, 2022–2023
- **Science and Technology Innovation Scholarship**, 2022–2023 and 2024–2025
- Sunshine Education Scholarship, Social Practice Scholarship, and Scholarship for Civic and Ethical Excellence, 2022–2023

SELECTED RESEARCH PROJECTS

Agent-based GEM Correction Ongoing

- Designing an evidence-aware workflow to identify candidate GEM errors and propose auditable corrections.
- Evaluation focuses on biochemical evidence, model behavior, and phenotype consistency rather than objective improvement alone.

Comparative GEM Reconstruction and Phenotype Analysis of Methylophilic Yeasts Undergraduate Thesis

- Reconstructed draft, strain-specific GEMs for 16 representative methylophilic yeasts using AlphaGEM and COBRApy; performed reference-guided gap filling for methanol growth.
- Combined FBA/pFBA with OrthoFinder-based orthology, gene-copy-number analysis, and methanol-utilization and tolerance phenotypes to study strain-level diversity.

TECHNICAL SKILLS

Computational biology: Genome-scale metabolic modeling, comparative genomics, phylogenomics, and genotype-phenotype analysis

Quantitative methods: Constraint-based modeling, FBA/pFBA, gap filling, GPR analysis, growth-curve modeling, and statistical analysis

Programming: Python, C++, MATLAB, and Git; reproducible research workflows, testing, debugging, and automation

RESEARCH INTERESTS

- Hybrid mechanistic-AI virtual cells
- Genome-scale metabolic modeling
- Multimodal cell-state and perturbation modeling
- Agent-based GEM correction